

Erdős Problem #477

EP-477

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Is there a polynomial $f : \mathbb{Z} \rightarrow \mathbb{Z}$ of degree at least 2 and a set $A \subset \mathbb{Z}$ such that for any $n \in \mathbb{Z}$ there is exactly one $a \in A$ and $b \in \{f(n) : n \in \mathbb{Z}\}$ such that $n = a + b$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>